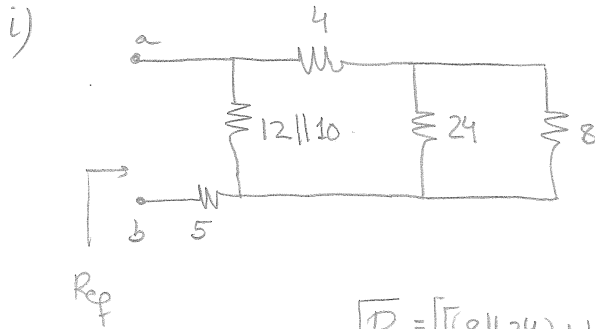
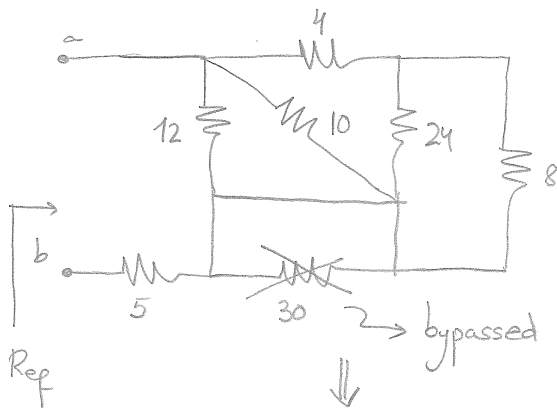


QUESTION 12



$$R_{eq} = \left[\left[(8 \parallel 24) + 4 \right] \parallel (12 \parallel 10) \right] + 5 =$$

$$= \left[(6 + 4) \parallel 5.4545 \right] + 5 = 3.53 + 5 = \boxed{8.53 \Omega}$$

ii)



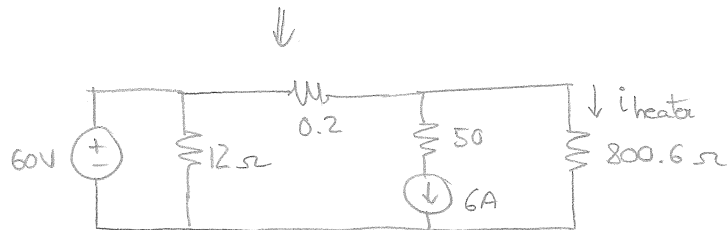
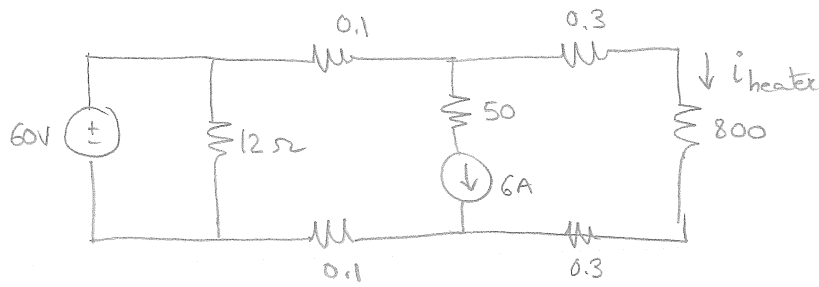
$$P_{R_{eq}} = \frac{V^2}{R} = \frac{30^2}{8.53} = 105.51 \text{ W}$$

$$P = \frac{\Delta w}{\Delta t} \Rightarrow w = P \times \Delta t = 105.51 \text{ W} \times 8 \text{ hr} \times 3600 \text{ s/hr}$$

$$= 3038686.99 \text{ J}$$

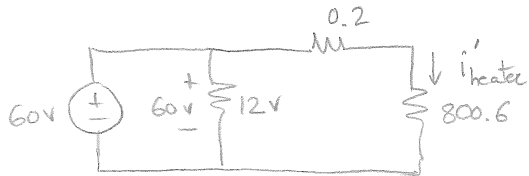
$$\approx 3038687 \text{ J} = 3.039 \text{ MJ}$$

QUESTION 1b



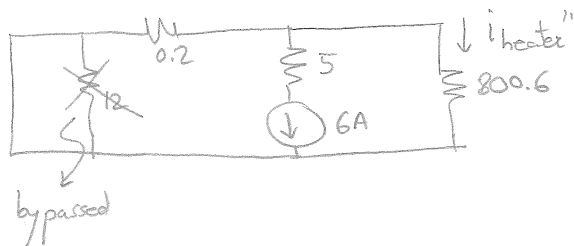
i)

* Contribution of voltage source



$$i'_{heater} = \frac{60}{0.2 + 800.6} = 0.075 \text{ A}$$

* Contribution of current source



$$i''_{heater} = -6 \frac{800.6 \parallel 0.2}{800.6} = -6 \times \frac{0.2}{800.6} = -0.0015 \text{ A}$$

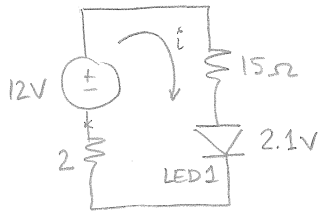
↑
current divider

* Add contributions

$$i_{heater} = i'_{heater} + i''_{heater} = 0.075 - 0.0015 = \underline{0.0735 \text{ A}}$$

QUESTION 2

i)

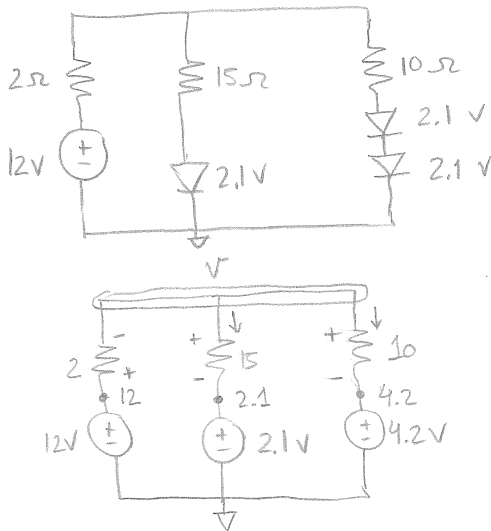


$$\text{KVL: } -12 + 15i + 2.1 + 2i = 0$$

$$-9.9 + 17i = 0 \Rightarrow i = \frac{9.9}{17} = 0.582 \text{ A} = i_{15\Omega}$$

$$P_{\text{LED1}} = V \times i = + 2.1 \times 0.582 = 1.222 \text{ W}$$

ii)



Use nodal to find V :

$$\text{KCL @ } V: \frac{12-V}{2} - \frac{V-2.1}{15} - \frac{V-4.2}{10} = 0$$

$$15(12-V) - 2(V-2.1) - 3(V-4.2) = 0$$

$$180 - 15V - 2V + 4.2 - 3V + 12.6 = 0$$

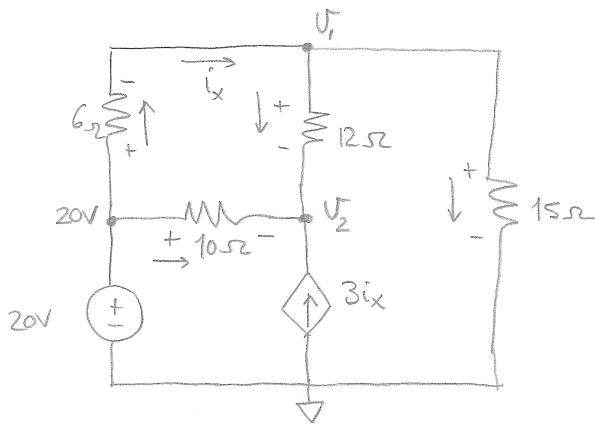
$$196.8 - 20V = 0 \Rightarrow V = \frac{196.8}{20}$$

$$V = 9.84 \text{ V}$$

$$i_{10\Omega} = \frac{V-4.2}{10} = \frac{9.84-4.2}{10} = 0.564 \text{ A}$$

$$P_{\text{LED2}} = P_{\text{LED3}} = V \times i_{10\Omega} = 2.1 \times 0.564 = 1.184 \text{ W}$$

QUESTION 3a



i)

$$\text{KCL @ } V_1: \underbrace{\frac{20 - V_1}{6}}_{i_x} - \frac{V_1 - V_2}{12} - \frac{V_1}{15} = 0 \quad (1)$$

$$\text{KCL @ } V_2: \frac{20 - V_2}{10} + 3 \frac{20 - V_1}{6} + \frac{V_1 - V_2}{12} = 0 \quad (2)$$

$$\frac{20}{6} - \left(\frac{1}{6} + \frac{1}{12} + \frac{1}{15} \right) V_1 + \frac{1}{12} V_2 = 0 \quad (\times 120)$$

$$2 + 10 + \left(-\frac{1}{2} + \frac{1}{12} \right) V_1 - \left(\frac{1}{10} + \frac{1}{12} \right) V_2 = 0 \quad (\times 60)$$

$$\begin{cases} 400 - 38V_1 + 10V_2 = 0 \\ 720 - 25V_1 - 11V_2 = 0 \end{cases} \Rightarrow \begin{cases} -418V_1 + 110V_2 = -4400 \\ -250V_1 - 110V_2 = -7200 \end{cases}$$

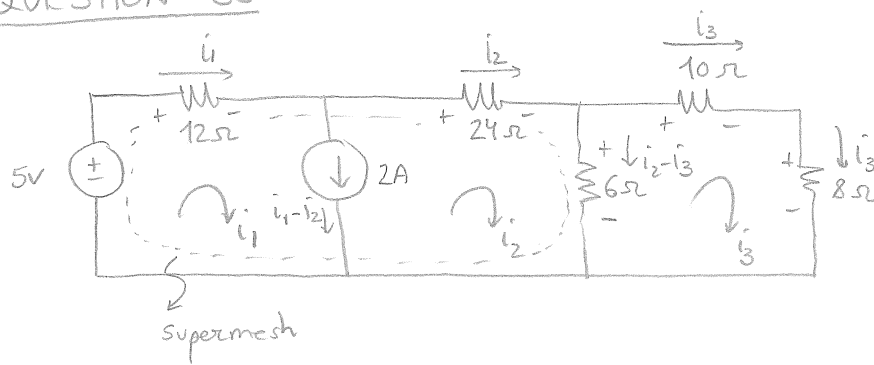
$$-668V_1 = -11600 \Rightarrow \boxed{V_1 = \frac{-11600}{-668} = 17.37V}$$

$$\boxed{V_2 = \frac{720 - 25V_1}{11} = \frac{720 - 25 \times 17.37}{11} = 25.98V}$$

ii)

$$\boxed{P_{6\Omega} = i_x^2 \times 6 = \left(\frac{20 - V_1}{6} \right)^2 \times 6 = \frac{(20 - 17.37)^2}{6} = 1.153W}$$

QUESTION 36



$$\text{KVL @ supermesh: } -5 + 12i_1 + 24i_2 + 6(i_2 - i_3) = 0$$

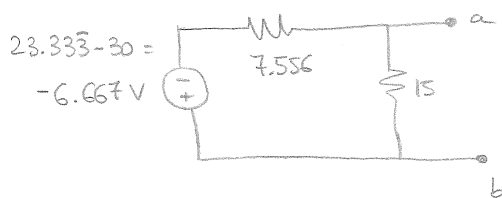
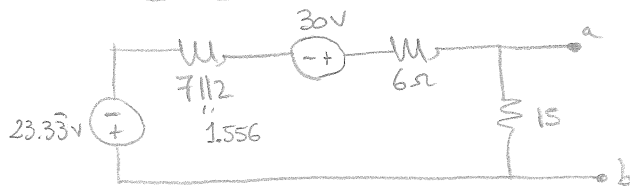
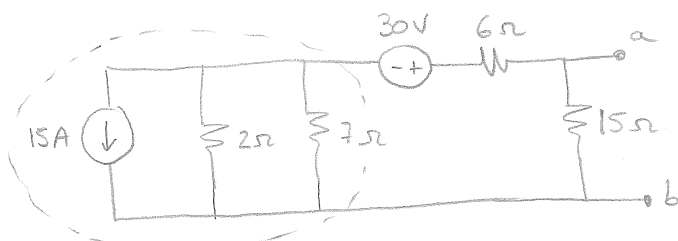
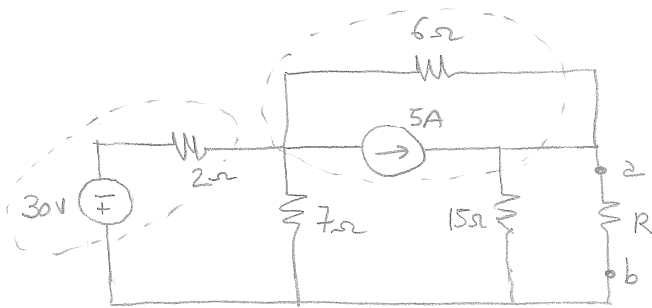
$$\text{KVL @ mesh 3: } -6(i_2 - i_3) + 10i_3 + 8i_3 = 0$$

$$\text{Constraint equation: } i_1 - i_2 = 2$$

Simplify:

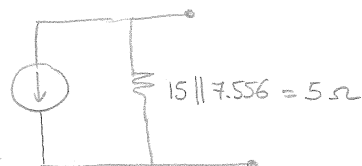
$$\left. \begin{aligned} 12i_1 + 30i_2 - 6i_3 &= 5 \\ -6i_2 + 24i_3 &= 0 \\ i_1 - i_2 &= 2 \end{aligned} \right\}$$

QUESTION 42



⇒

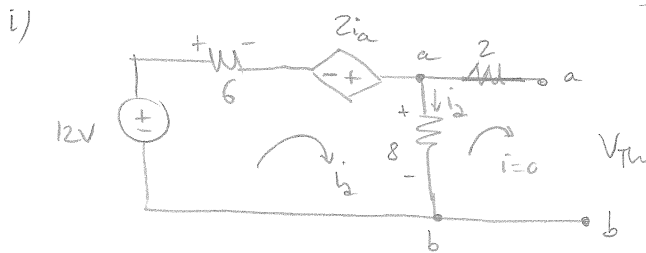
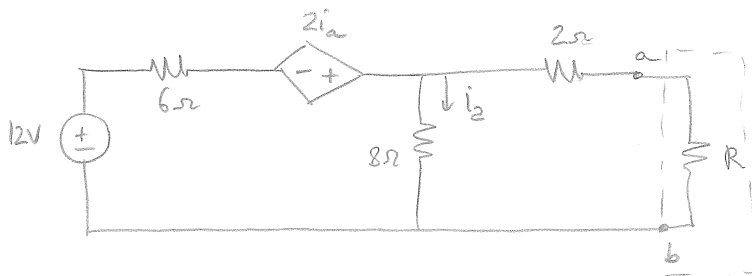
$$\frac{-6.667}{7.556} = -0.88A$$



$$I_N = -0.88A$$

$$R_N = 5\Omega$$

QUESTION 46



$$V_{th} = V_{oc} = V_{ab} = V_{8\Omega}$$

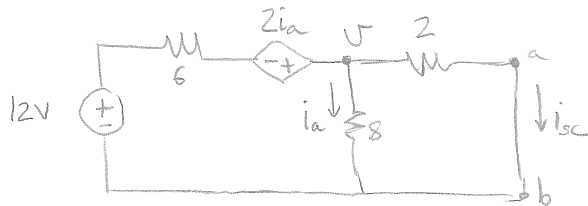
$$\text{KVL @ mesh: } -12 + 6i_a - 2i_a + 8i_a = 0$$

$$-12 + 12i_a = 0 \Rightarrow i_a = \frac{12}{12} = 1A$$

$$\boxed{V_{th} = V_{8\Omega} = 8i_a = 8V}$$

ii) Calculate R_{th} using method 2:

$$R_{th} = \frac{V_{th}}{i_{sc}}$$



$$i_a = \frac{V}{8}$$

$$V = (12 + 2i_a) \frac{2 \parallel 8}{6 + 2 \parallel 8} = \left(12 + \frac{V}{4}\right) \frac{1.6}{7.6} = 0.21 \left(12 + \frac{V}{4}\right)$$

$$4V = 0.21(48 + V)$$

$$4V - 0.21V = 0.21 \times 48$$

$$3.79V = 10.08$$

$$V = \frac{10.08}{3.79} = 2.66V$$

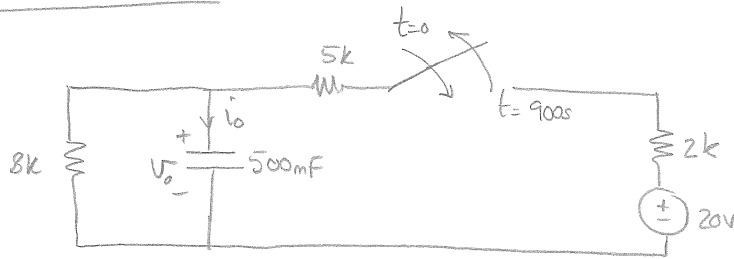
$$i_{sc} = \frac{V}{2} = \frac{2.66}{2} = 1.33A$$

$$\Rightarrow \boxed{R_{th} = \frac{V_{th}}{i_{sc}} = \frac{8V}{1.33A} = 6\Omega}$$

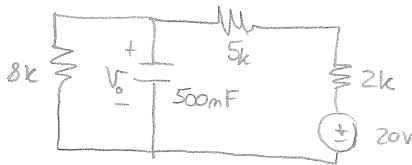
iii)

$$\boxed{P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{8^2}{4 \times 6} = 2.67W}$$

QUESTION 5



i) $0 \leq t < 900\text{s}$
 $\hookrightarrow$ switch closed

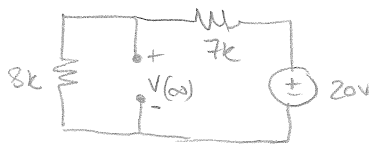


Step response

$$V_o = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}, \text{ where } \tau = RC$$

* Calculate $V(0)$ for $t < 0 \Rightarrow \underline{V(0) = 0\text{V}}$

* Calculate $V(\infty)$ for $0 \leq t < 900\text{s}$



$$\underline{V(\infty) = V_{8k} = 20 \frac{8k}{7k + 8k} = \frac{20 \times 8}{7+8} = 10.667\text{V}}$$

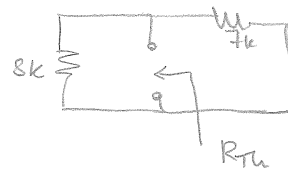
$\uparrow$
voltage divider

* Calculate τ :

$$\tau = R_{Th} C$$

$\rightarrow$ Calculate R_{Th} (method 1):

$$\tau = R_{Th} C = 3.733k \times 500\mu\text{F} = 1866.67\text{s}$$



$$R_{Th} = 8k \parallel 7k = 3.733\text{k}\Omega$$

* Substitute:

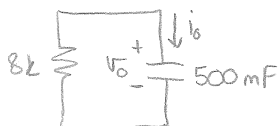
$$\boxed{V_o = 10.667 + [0 - 10.667]e^{-t/1866.67} = 10.667(1 - e^{-t/1866.67})\text{V}}$$

ii) $W = \frac{1}{2} CV^2 \Rightarrow \boxed{W = \frac{1}{2} \times 0.5 \times V_o^2(300\text{s}) = 0.25 \times 1.58^2 = 0.627\text{J}}$

$$V_o(300\text{s}) = 10.667(1 - e^{-300/1866.67}) = 1.58\text{V}$$

iii) $t \geq 900\text{s}$

$\hookrightarrow$ switch open



Natural response

$$V_o = V(900\text{s}) e^{-(t-900)/\tau}, \text{ where } \tau = RC$$

* $V(900\text{s}) = 10.667(1 - e^{-900/1866.67}) = 4.08\text{V}$
 $\uparrow$
 from previous interval

* $\tau = RC = 8k \times 500\mu\text{F} = 4000\text{s}$

$$V_o = 4.08 e^{-(t-900)/4000}$$

$$\hookrightarrow i_o = C \frac{dV_o}{dt} = 0.5 \frac{d}{dt} [4.08 e^{-(t-900)/4000}] = 0.5 \times 4.08 \times \left(\frac{-1}{4000}\right) e^{-(t-900)/4000}$$

$$\boxed{i_o = -0.00051 e^{-(t-900)/4000}\text{A}}$$